Van Hove function

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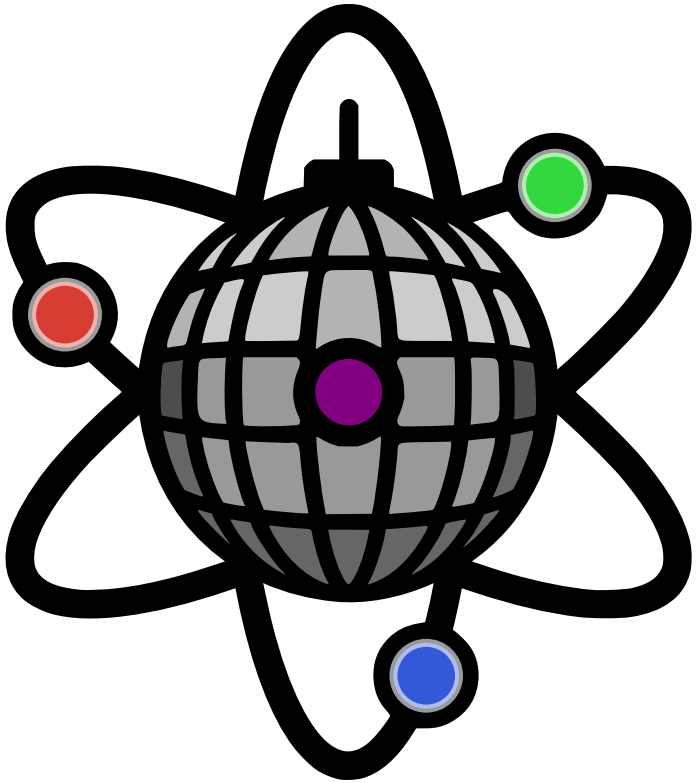
MDANSE Workshop 2026

04/11/2026



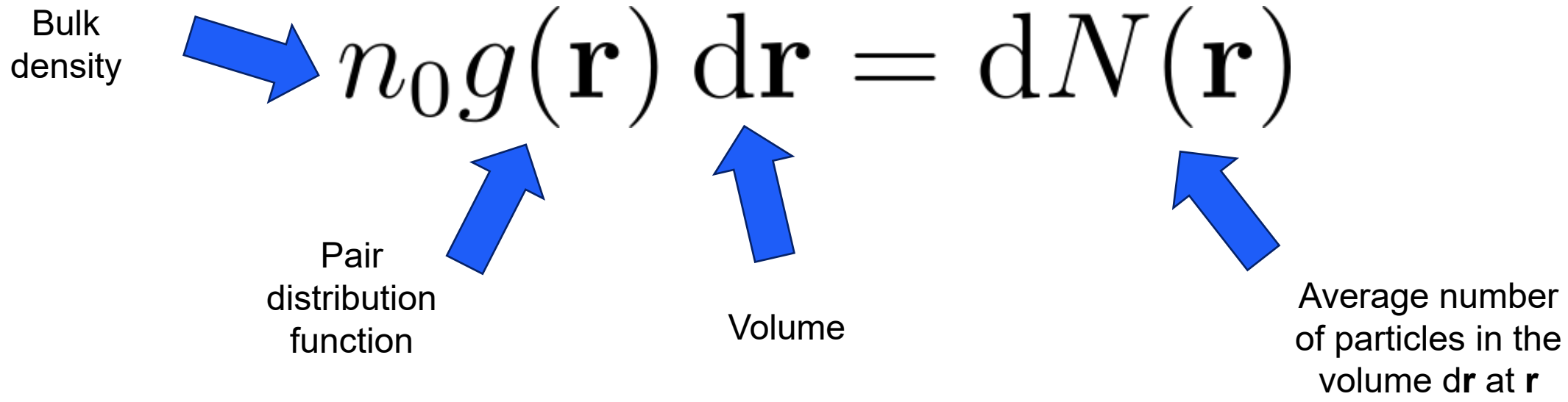
Van Hove function

- In this workshop exercise, we will run the van Hove calculations with MDANSE on our OpenMM water trajectory.
- But what is the van Hove function?



Pair Distribution Function

- Let's start with the pair distribution function which provides us with some information about how atoms are distributed in the system.
- It tells us what the average number of particles will be at a distance **r** in volume d**r** from an atom.



Pair Distribution Function

- Let's start with the pair distribution function which provides us with some information about how atoms are distributed in the system.
- In this case the pair distribution function tells us the average number of particles in the shell volume $4\pi r^2\,dr$ from a distance r of an atom.

Bulk density

$$n_0 g(r)\, 4\pi r^2\, \mathrm{d}r = \mathrm{d}N(r)$$

Pair distribution function

Shell volume

Average number of particles in the shell between r and r + dr

Pair Distribution Function

- The PDF can be written as a sum of delta functions.

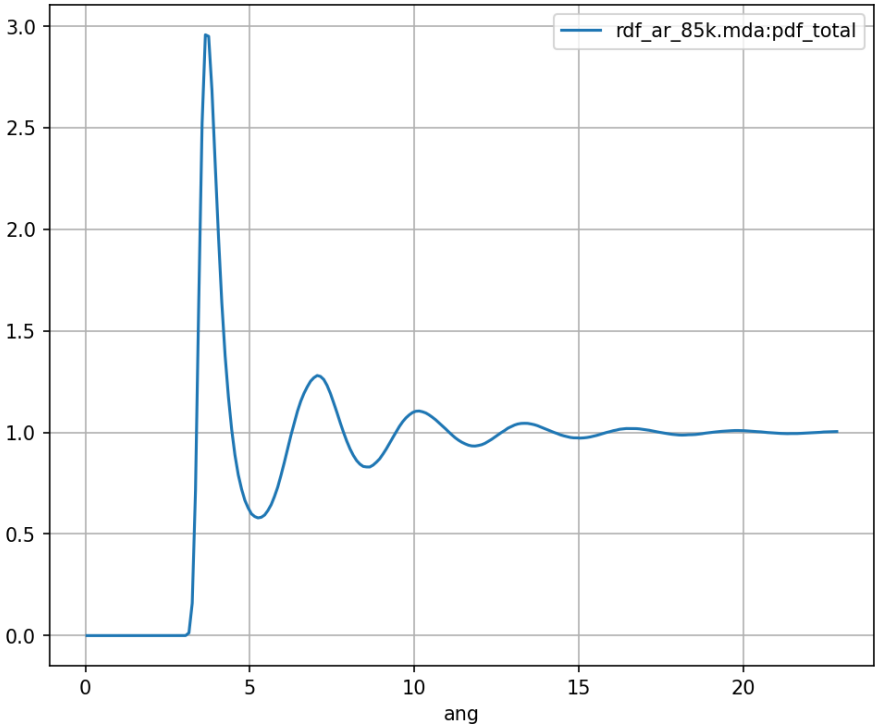
$$n_0 g(r)\, 4\pi r^2\, \mathrm{d}r = \mathrm{d}N(r)$$

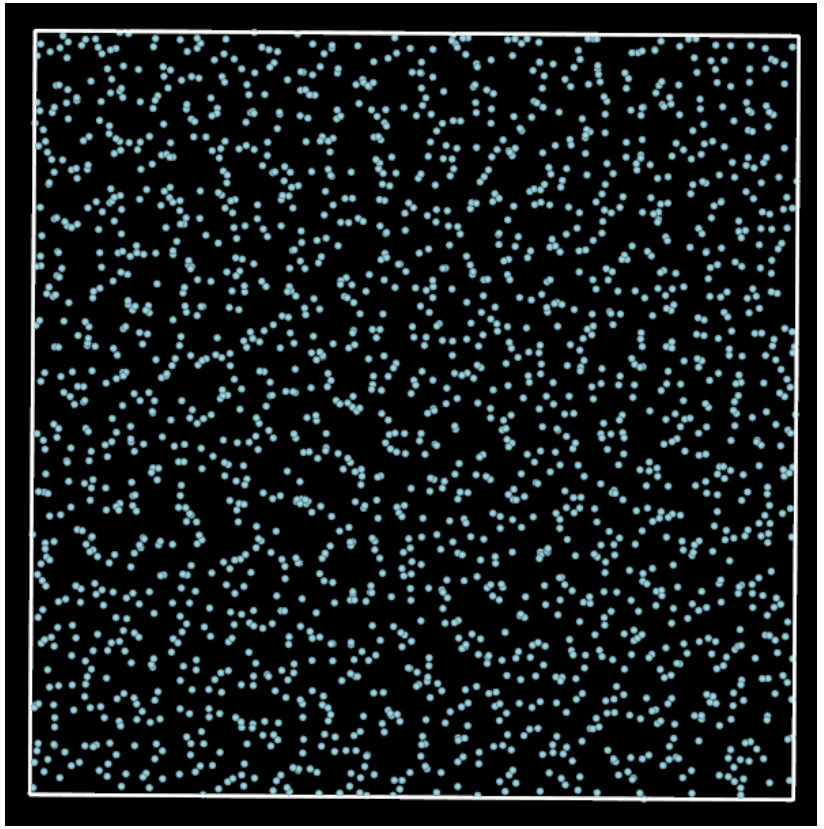
$$n_0 g(r) = \frac{1}{4\pi r^2}\frac{1}{N}\sum_{k\neq j}\left\langle \delta[r - |\mathbf{r}_k(t_0) - \mathbf{r}_j(t_0)|]\right\rangle_{t_0}$$

Pair Distribution Function

- Here is the PDF of a liquid argon trajectory generated from MDANSE.

This Ar trajectory was generated using lammps at 85 K with the Lennard-Jones potential.

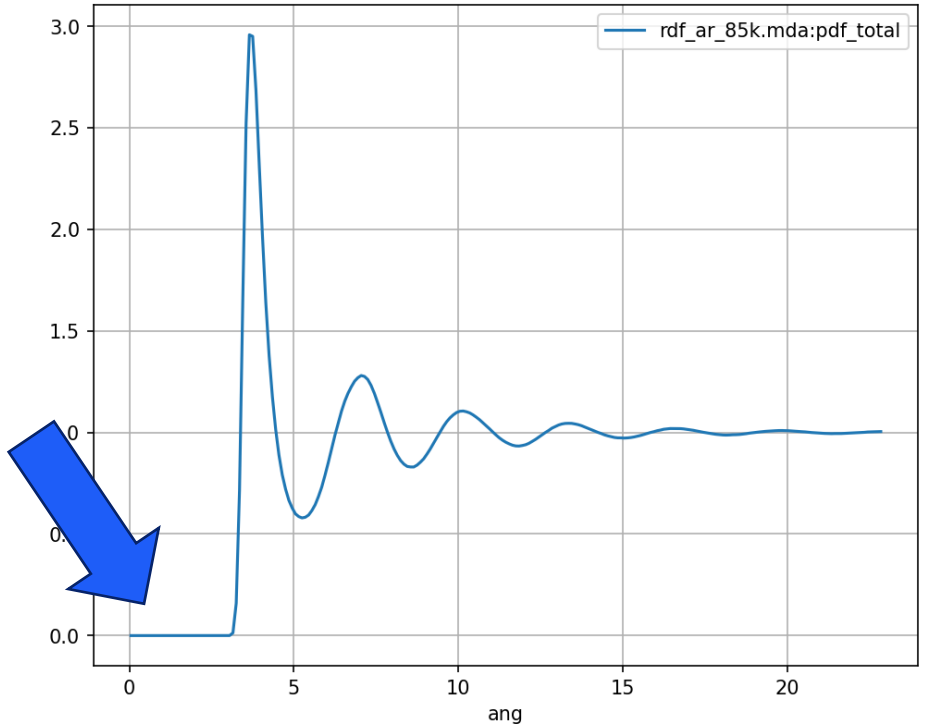


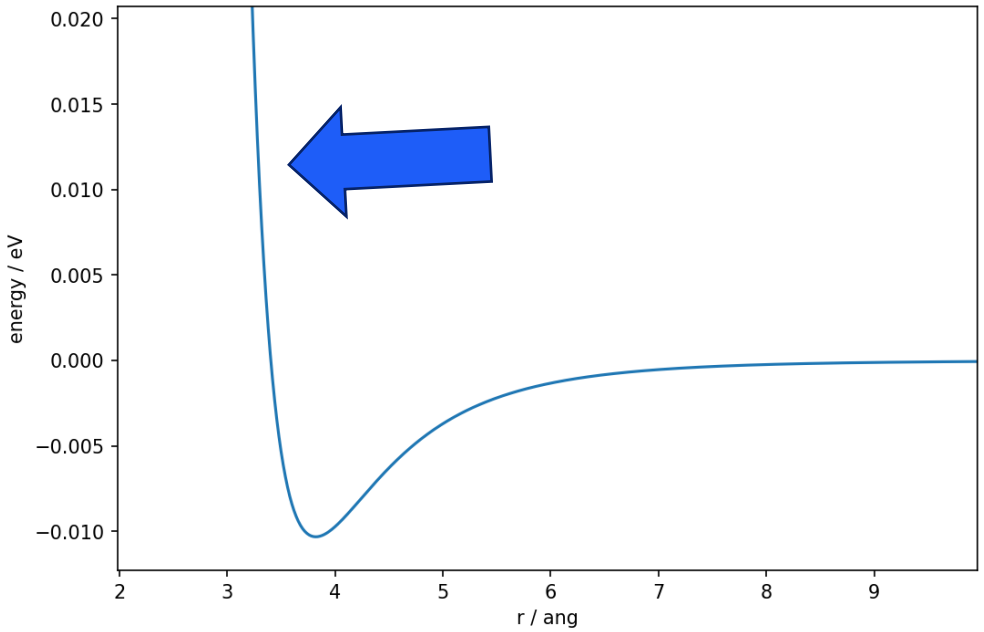


Pair Distribution Function

- Here is the PDF of a liquid argon trajectory generated from MDANSE.

We can see the repulsive $1/r^{12}$ potential causing a reduction of particles close to each atom.

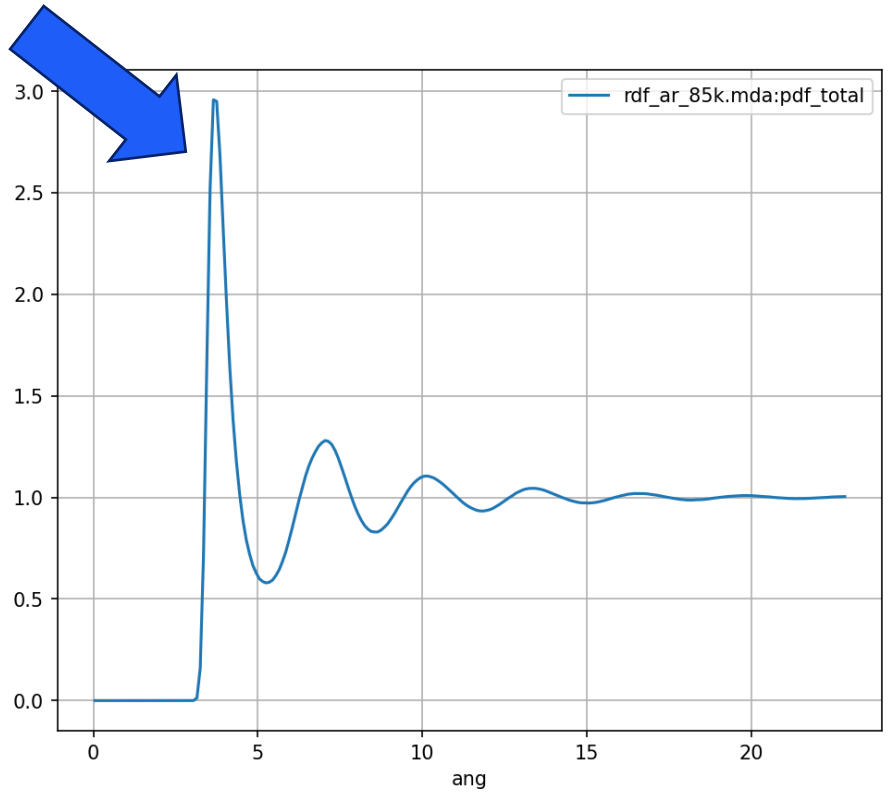


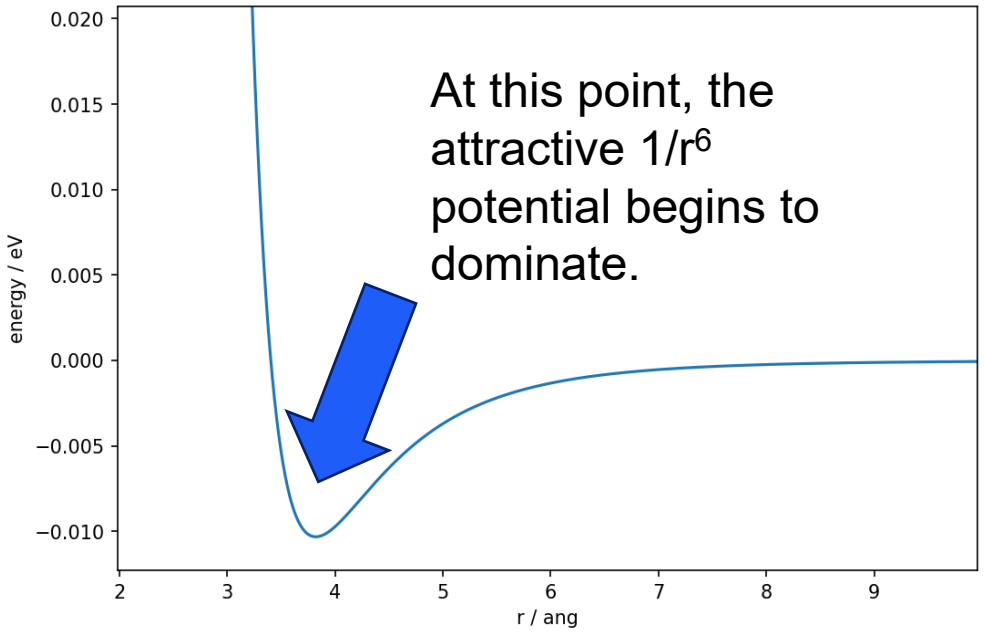


Pair Distribution Function

- Here is the PDF of a liquid argon trajectory generated from MDANSE.

The first peak in the PDF occurs around 3.7 Å this is approximately the minimum of the Lennard-Jones potential 3.8 Å.

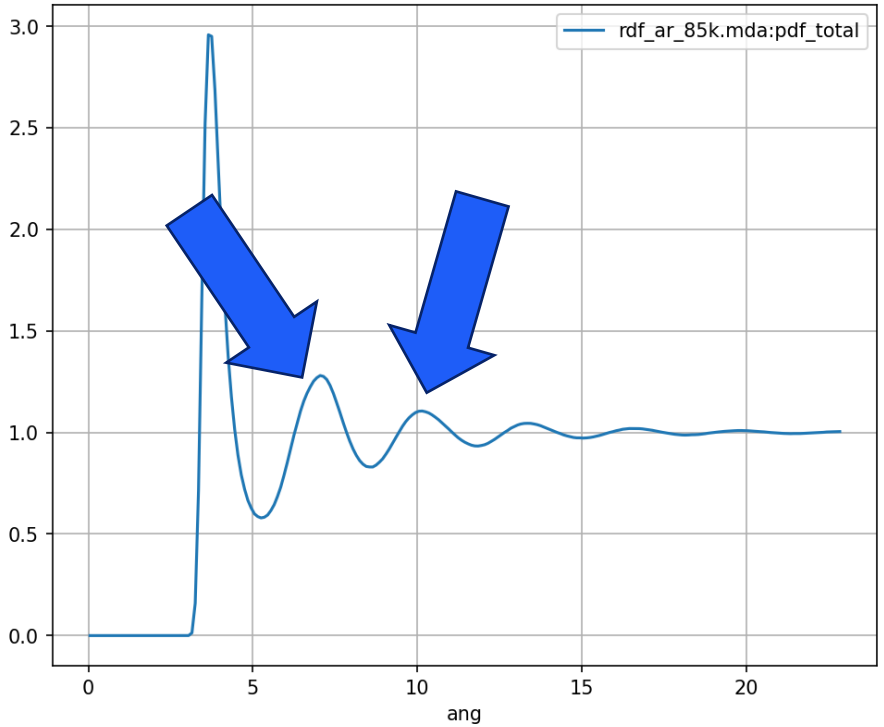


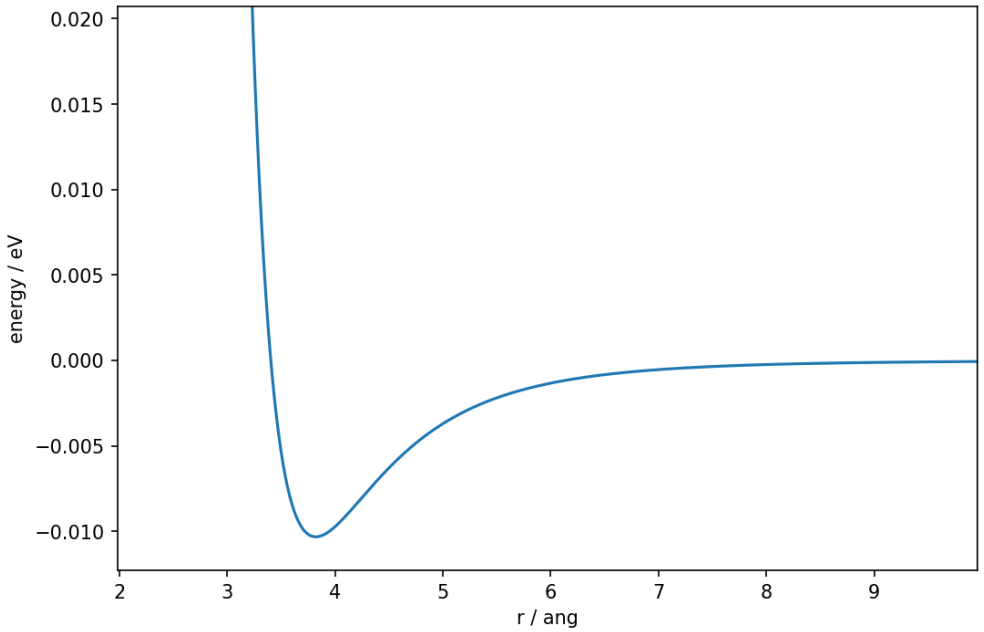


Pair Distribution Function

- Here is the PDF of a liquid argon trajectory generated from MDANSE.

The other peaks are the 2nd, 3rd and other shell of atoms around each atoms.

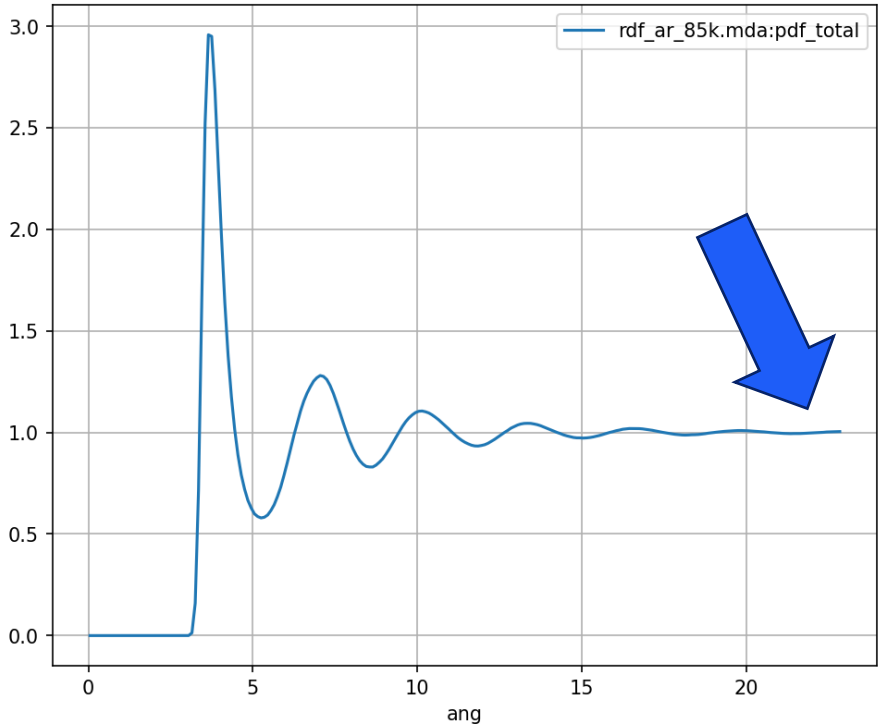


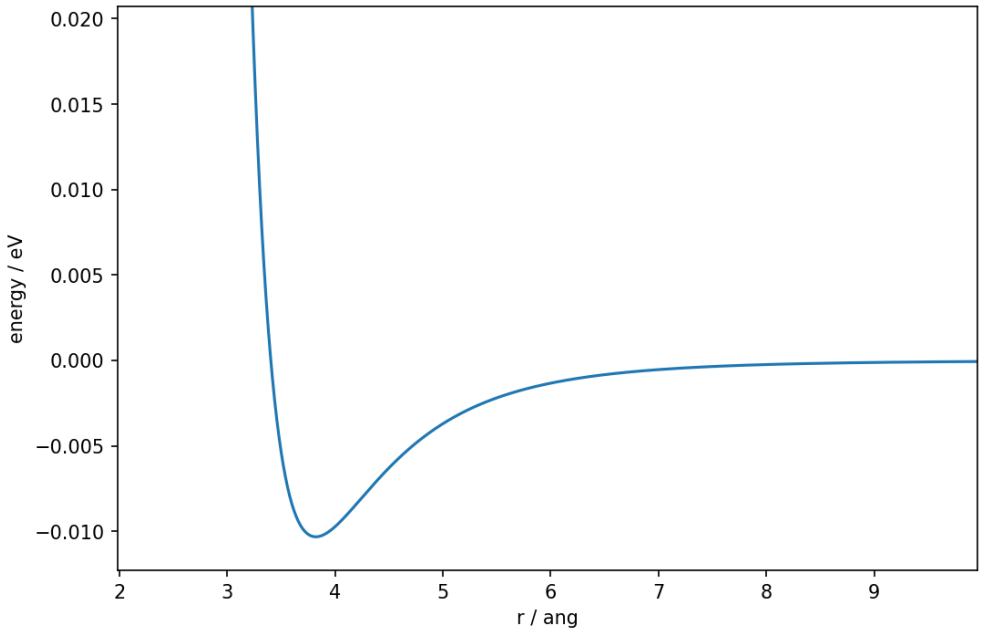


Pair Distribution Function

- Here is the PDF of a liquid argon trajectory generated from MDANSE.

For gas and liquid systems for large distance the PDF is normalized so that it goes to 1.

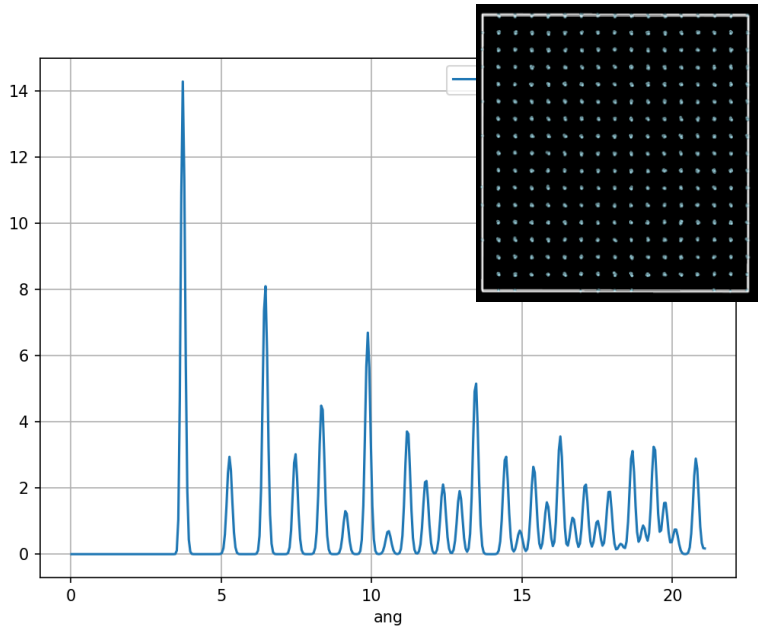




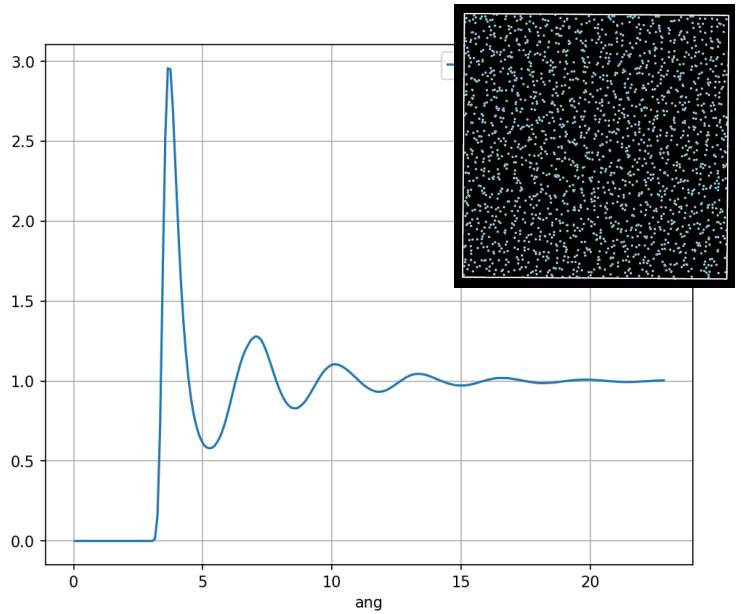
Pair Distribution Function

- The PDF can be quite different when the system exist in a different phase.

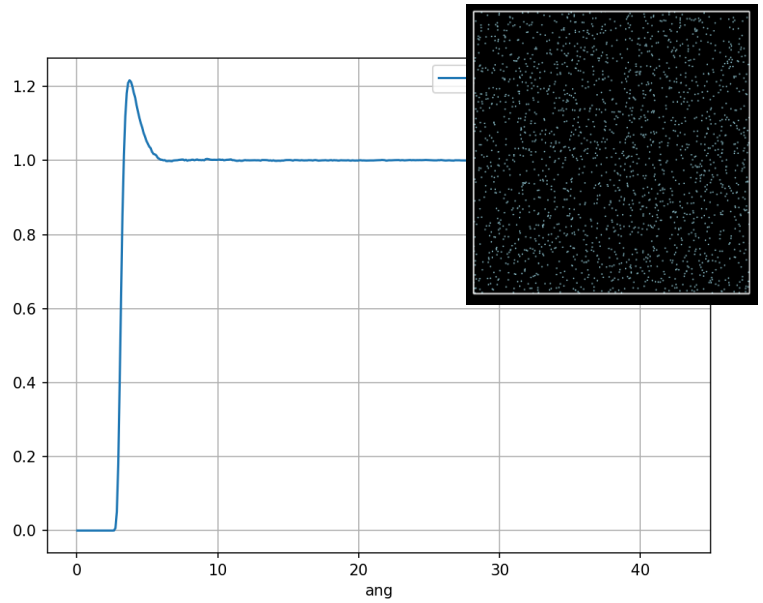
Solid (ordered)



Liquid



Gas



Van Hove Function

$$n_0 g(\mathbf{r}) = \frac{1}{N} \sum_{j \neq k} \langle \delta[\mathbf{r} - \mathbf{r}_k(t_0) + \mathbf{r}_j(t_0)] \rangle_{t_0}$$

- Similarly to the PDF the van Hove function is also a sum of delta function.

$$G(\mathbf{r}, t) = \frac{1}{N} \sum_{jk} \langle \delta[\mathbf{r} - \mathbf{r}_k(t_0 + t) + \mathbf{r}_j(t_0)] \rangle_{t_0}$$

- The PDF gives use some insights into the structure of our system while the van Hove function gives us insights into the structure and dynamics of the system.

Van Hove Function

- The van Hove function can be split in self and distinct terms

$$G_{\mathrm{s}}(\mathbf{r},t) = \frac{1}{N}\sum_{j}\left\langle \delta\left[\mathbf{r} - \mathbf{r}_j(t_0+t) + \mathbf{r}_j(t_0)\right]\right\rangle_{t_0}$$

$$G_{\mathrm{d}}(\mathbf{r},t) = \frac{1}{N}\sum_{j\neq k}\left\langle \delta\left[\mathbf{r} - \mathbf{r}_k(t_0+t) + \mathbf{r}_j(t_0)\right]\right\rangle_{t_0}$$

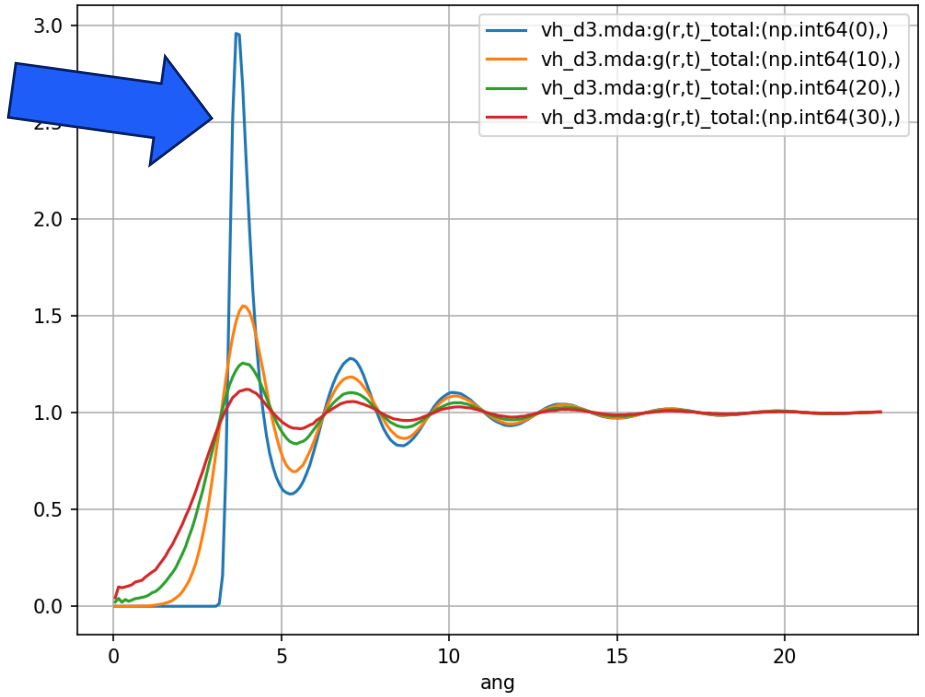
Van Hove Function (distinct-part)

- The distinct-part of the van Hove function can be written so that it is only a function of the distance r.

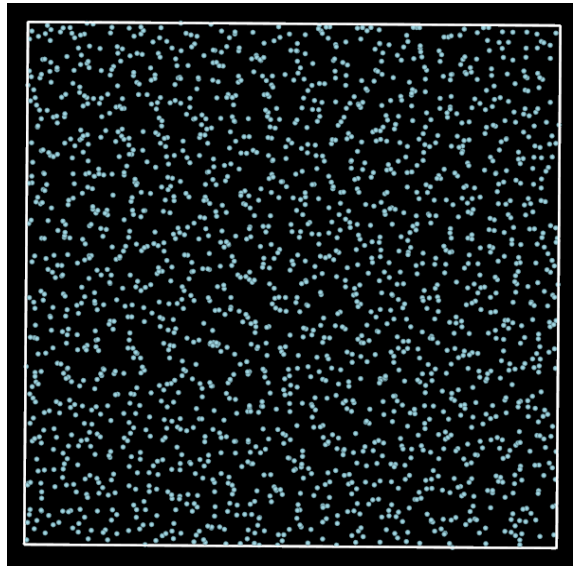
$$G_{\mathrm{d}}(r,t) = \frac{1}{4\pi r^2}\frac{1}{N}\sum_{j\neq k}\langle\delta[r-|\mathbf{r}_k(t_0+t)-\mathbf{r}_j(t_0)|]\rangle_{t_0}$$

At t=0 the distinct-part of the van Hove function is the PDF.

$$G_{\mathrm{d}}(\mathbf{r},0) = n_0 g(\mathbf{r})$$

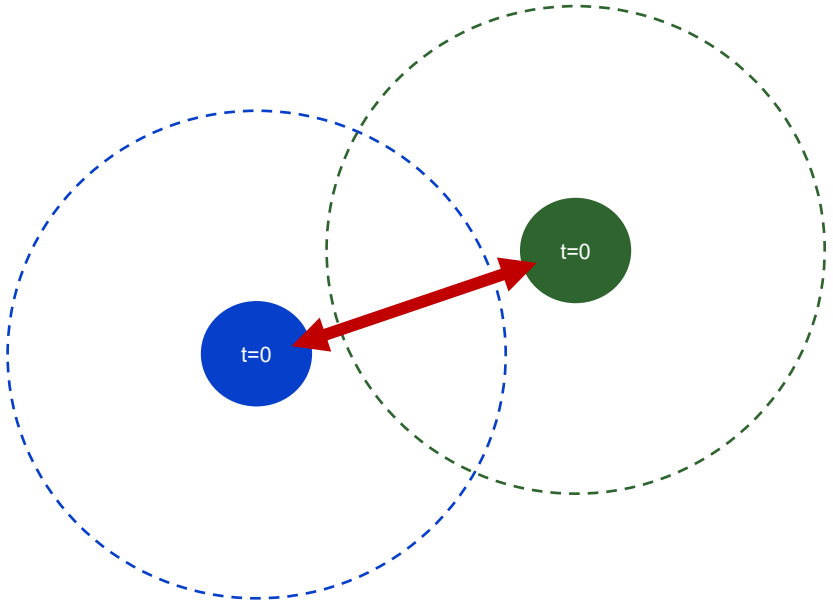


This is the liquid Ar MD trajectory at 85K.

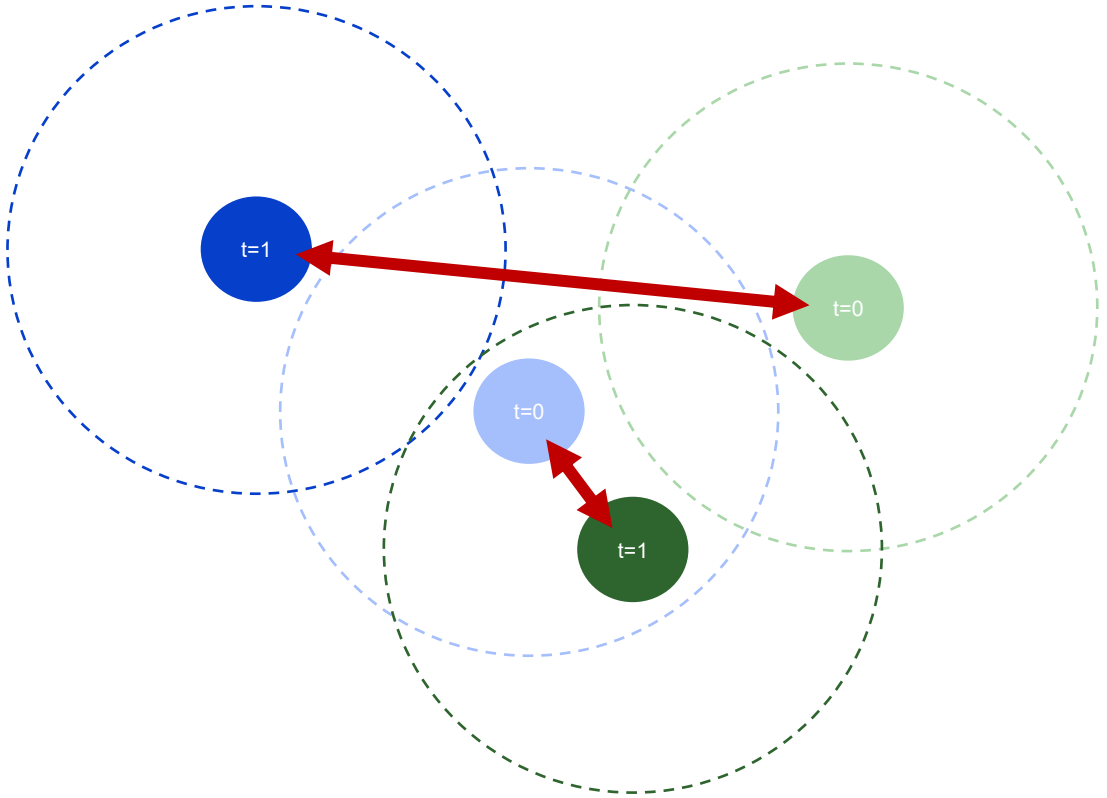


Van Hove Function (distinct-part)

$$G_{\mathrm{d}}(\mathbf{r}, t = 0)$$



$$G_{\mathrm{d}}(\mathbf{r}, t = 1)$$

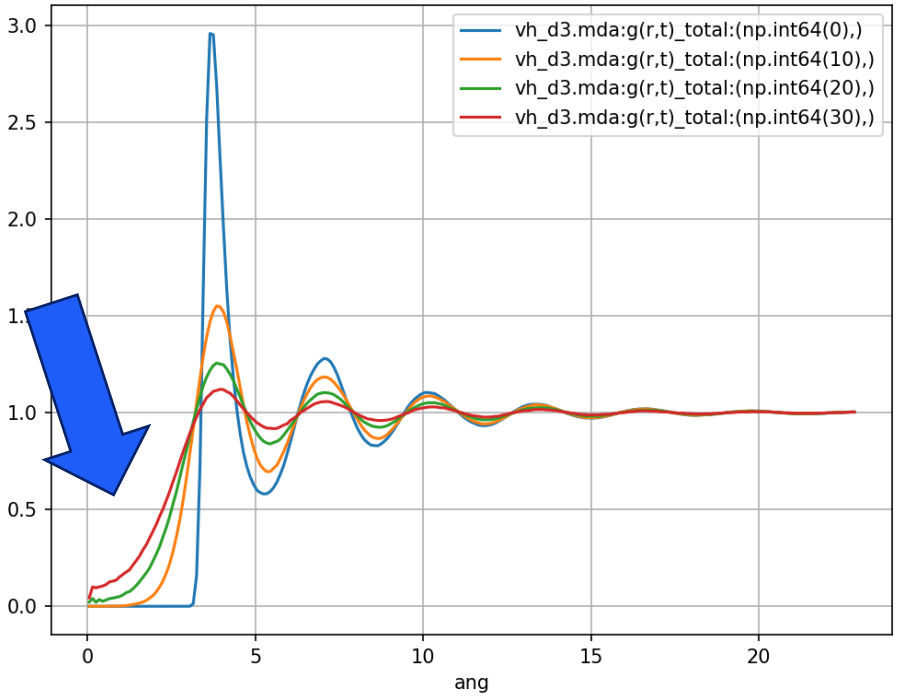


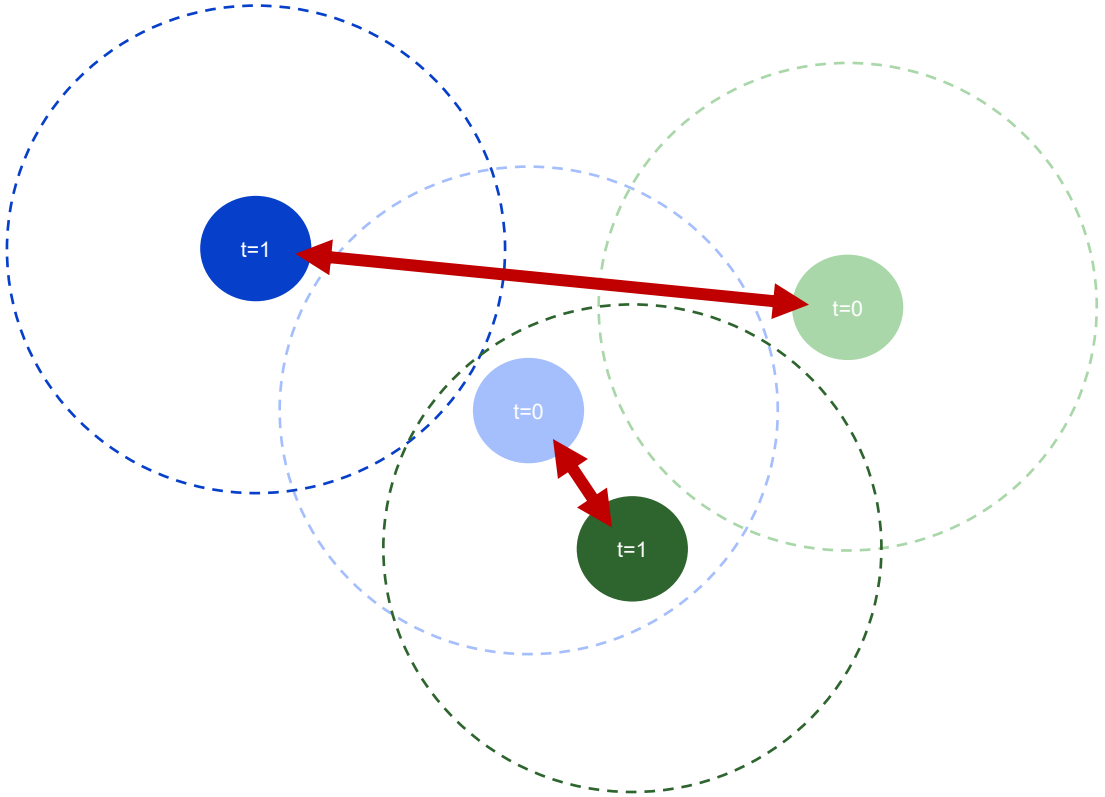
Van Hove Function (distinct-part)

- The distinct-part of the van Hove function can be written so that it is only a function of the distance r.

$$G_{\mathrm{d}}(r,t) = \frac{1}{4\pi r^2}\frac{1}{N}\sum_{j\neq k}\langle\delta[r-|\mathbf{r}_k(t_0+t)-\mathbf{r}_j(t_0)|]\rangle_{t_0}$$

As time progress the particle (j) moves away from its initial position and the hole that was around that particle begin to be occupied by other atoms.





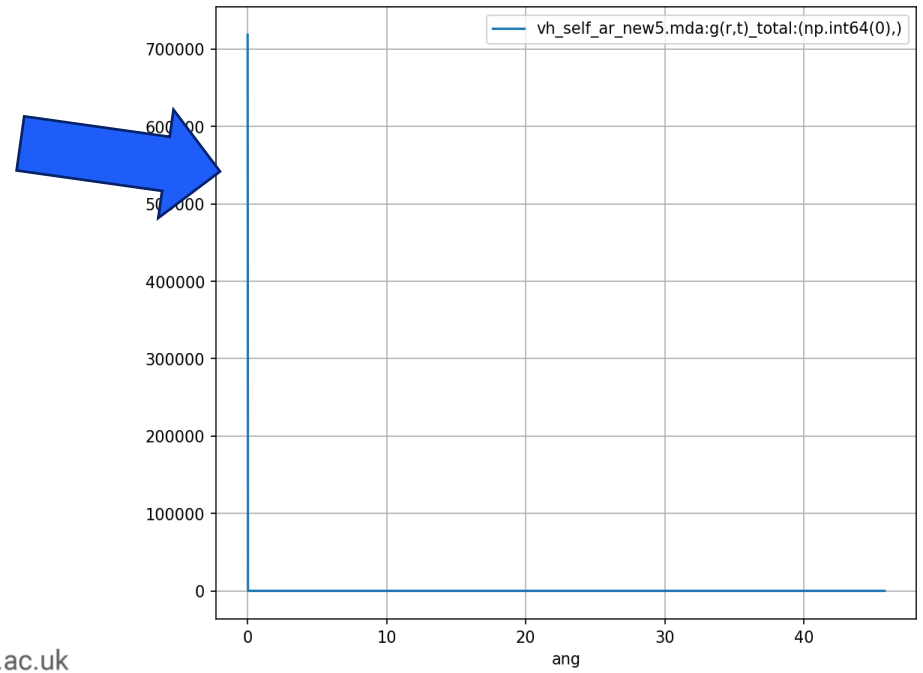
Van Hove Function (self-part)

- The self-part of the van Hove function can be written so that it is only a function of the distance r.

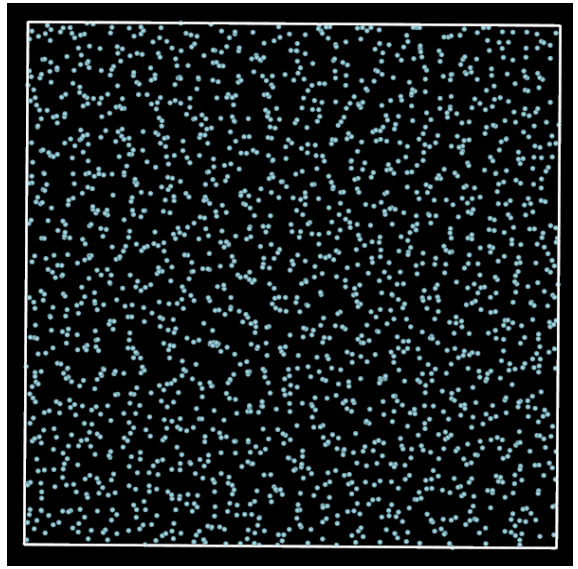
$$G_s(r,t) = \frac{1}{4\pi r^2}\frac{1}{N}\sum_j \langle \delta[r - |\mathbf{r}_j(t_0+t) - \mathbf{r}_j(t_0)|]\rangle_{t_0}$$

At t=0 the self-part of the van Hove function is a delta function.

$$G_s(\mathbf{r},0) = \delta(\mathbf{r})$$

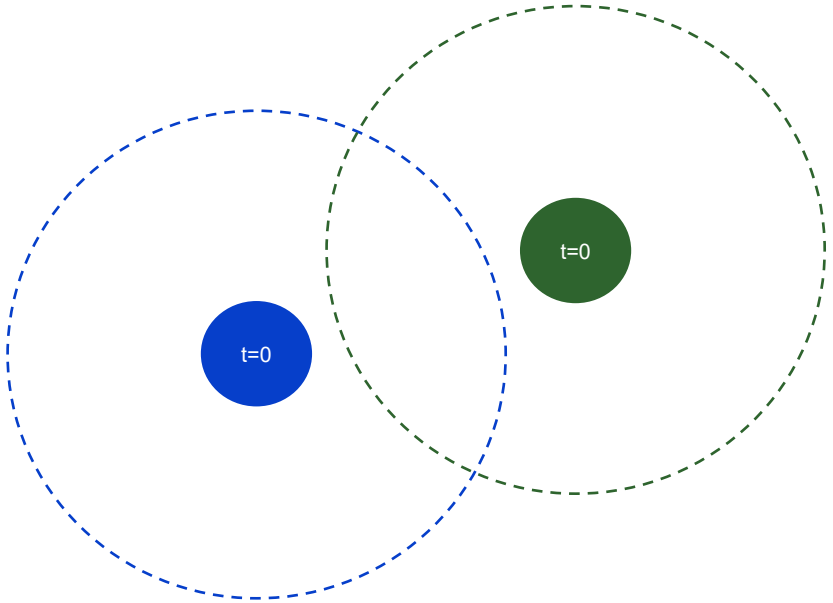


This is the liquid Ar MD trajectory at 85K.

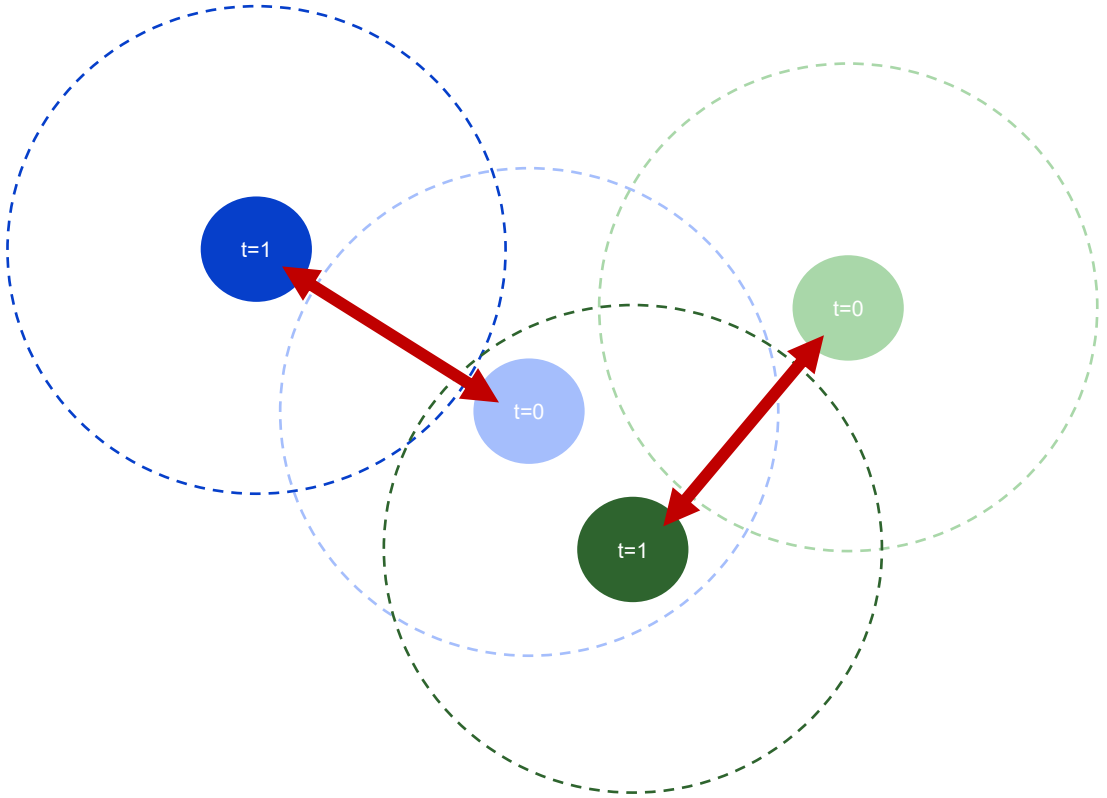


Van Hove Function (self-part)

$$G_s(\mathbf{r}, t = 0)$$



$$G_s(\mathbf{r}, t = 1)$$

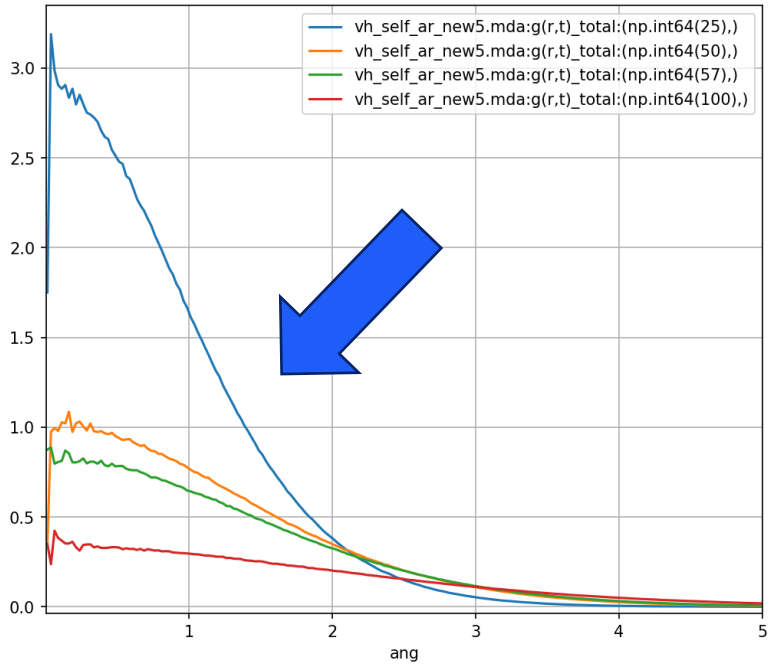


Van Hove Function (self-part)

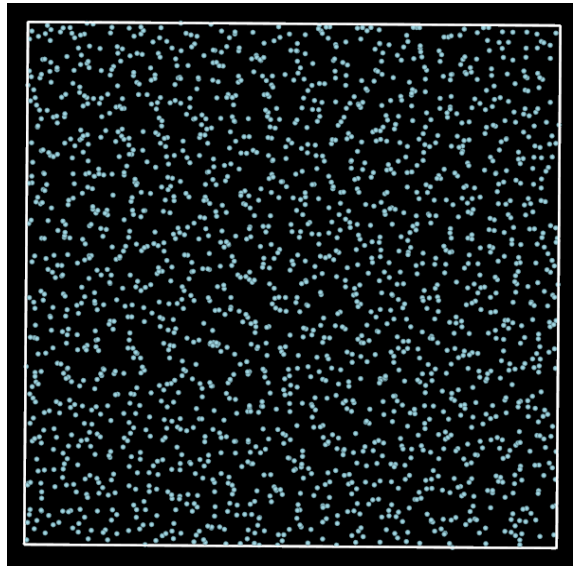
- The self-part of the van Hove function can be written so that it is only a function of the distance r.

$$G_s(r,t) = \frac{1}{4\pi r^2}\frac{1}{N}\sum_j \langle \delta[r - |\mathbf{r}_j(t_0+t) - \mathbf{r}_j(t_0)|]\rangle_{t_0}$$

The Ar atoms diffuse away from its initial position. The self-part of the van Hove function moves and spreads out as time progresses.



This is the liquid Ar MD trajectory at 85K.

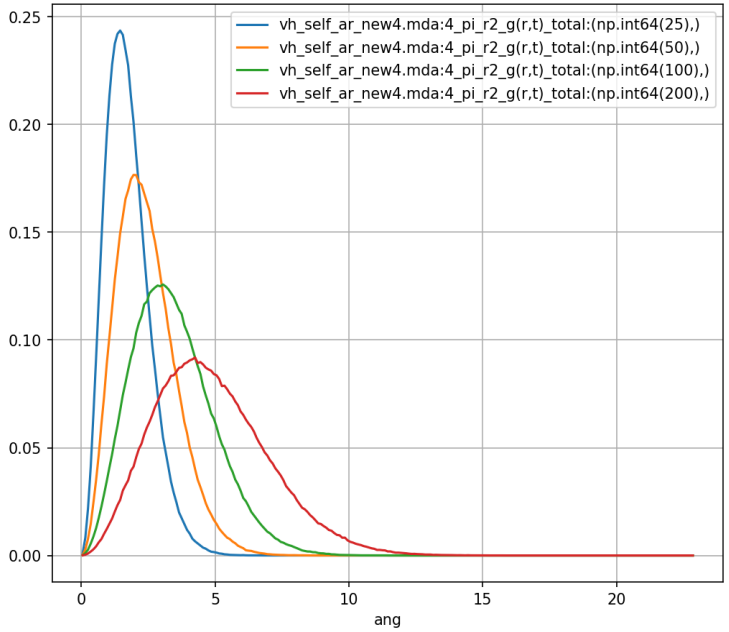


Van Hove Function (self-part)

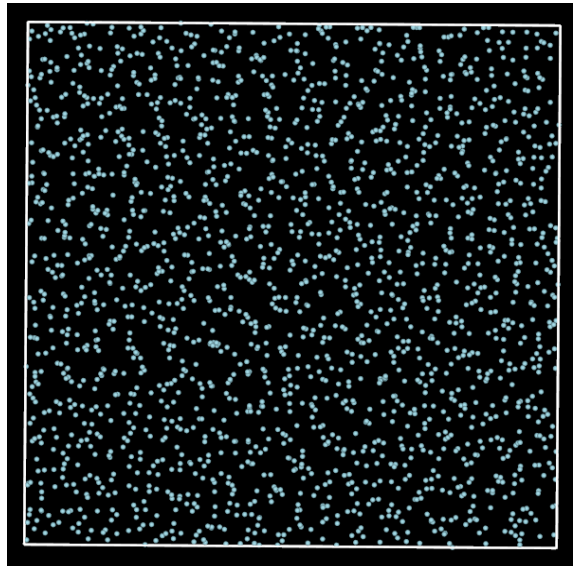
- The self-part of the van Hove function can be written so that it is only a function of the distance r.

$$4\pi r^2 G_s(r,t) = \frac{1}{N}\sum_j \langle \delta[r - |\mathbf{r}_j(t_0+t) - \mathbf{r}_j(t_0)|] \rangle_{t_0}$$

It is also useful to plot a normalized self-part of the van Hove function.



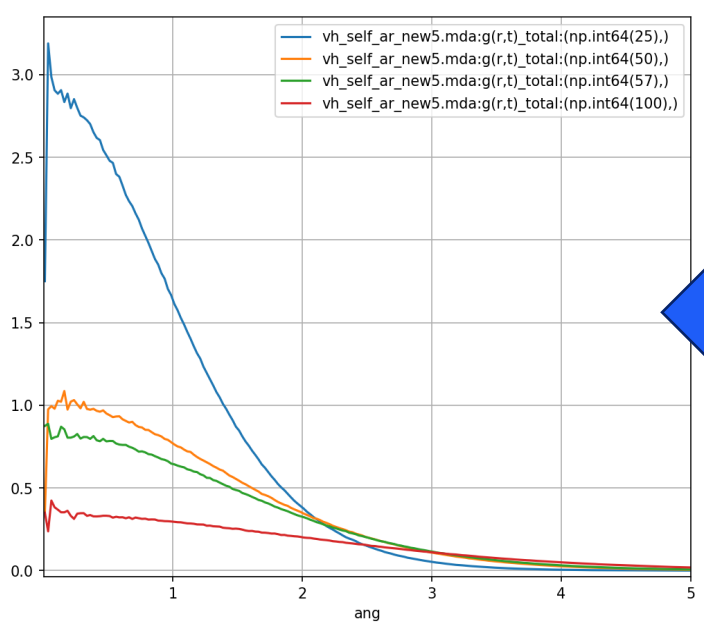
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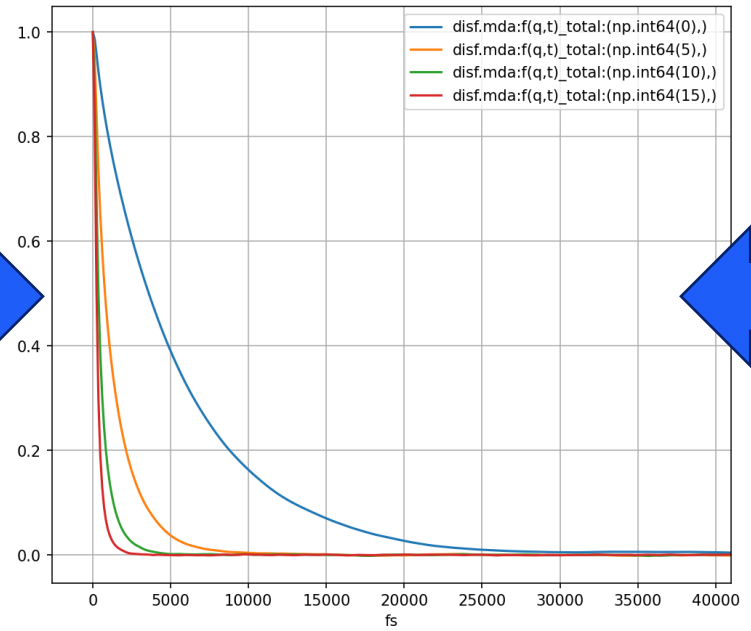


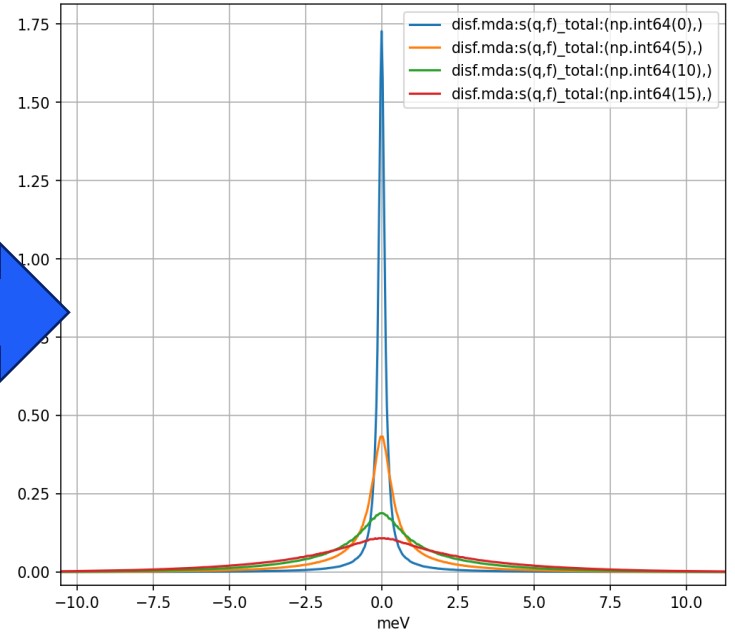
Van Hove Function (relation to neutron scattering)

- The van Hove function (and therefore the PDF) are closely related to the dynamics structure factor via Fourier transforms. Here we show the self-part and its Fourier transforms.

$G_s(\mathbf{r}, t)$ $F_s(\mathbf{Q}, t)$ $S_i(\mathbf{Q}, \omega)$

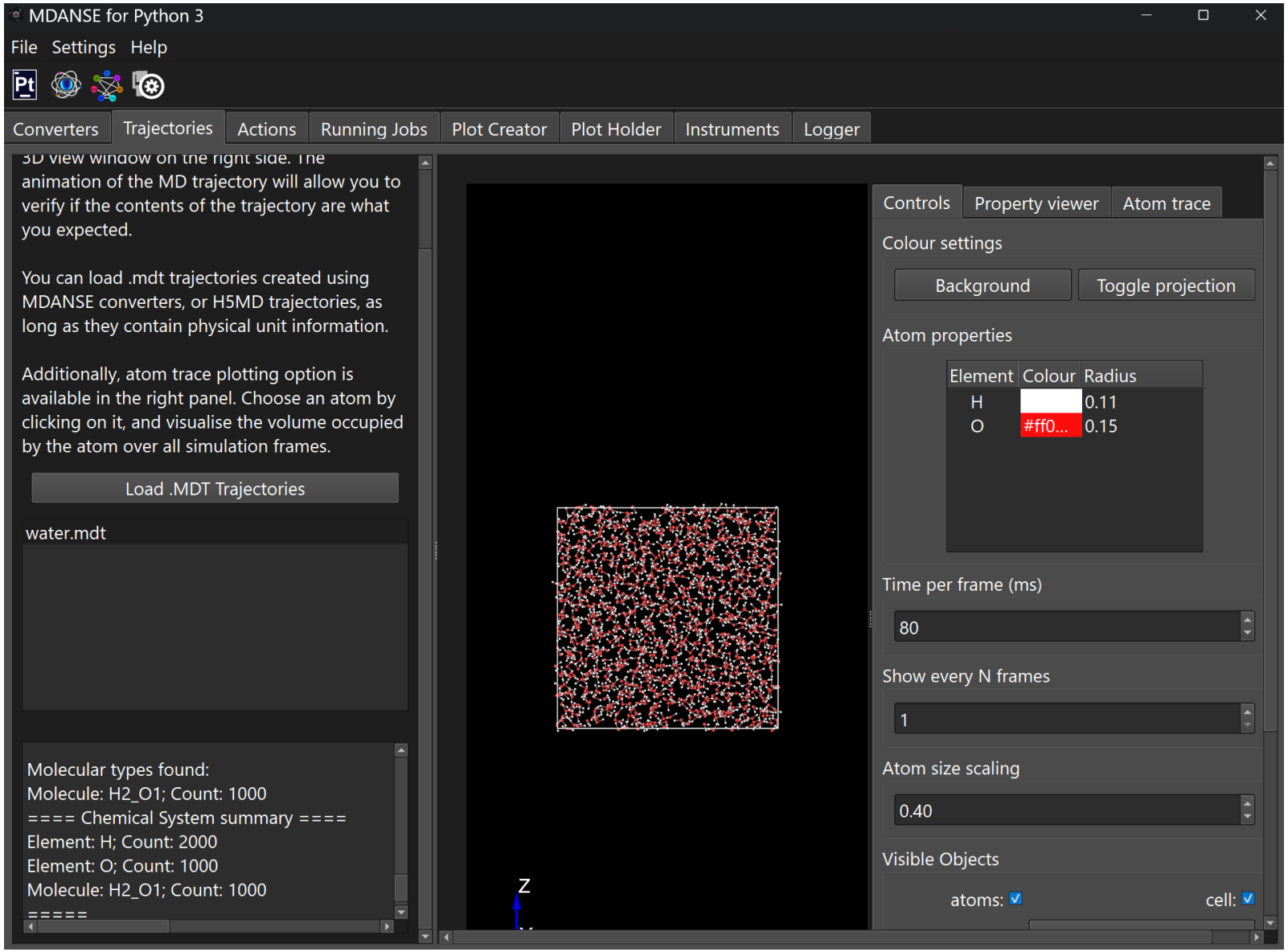






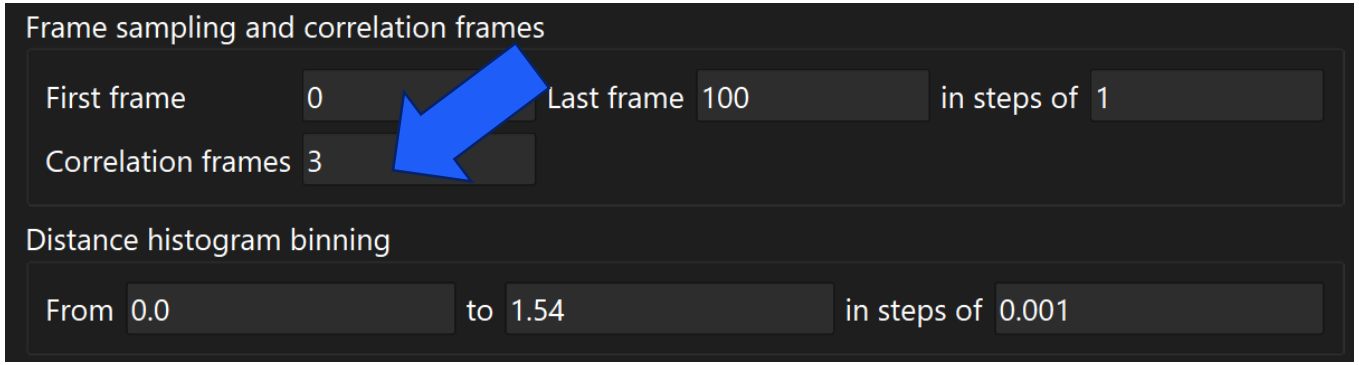
MDANSE

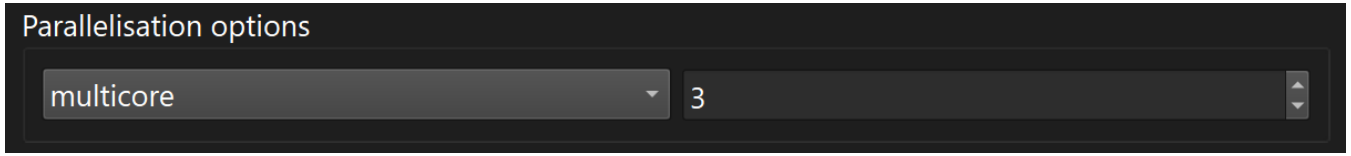
- Load the water trajectory that you created previously in the MDANSE GUI.



MDANSE

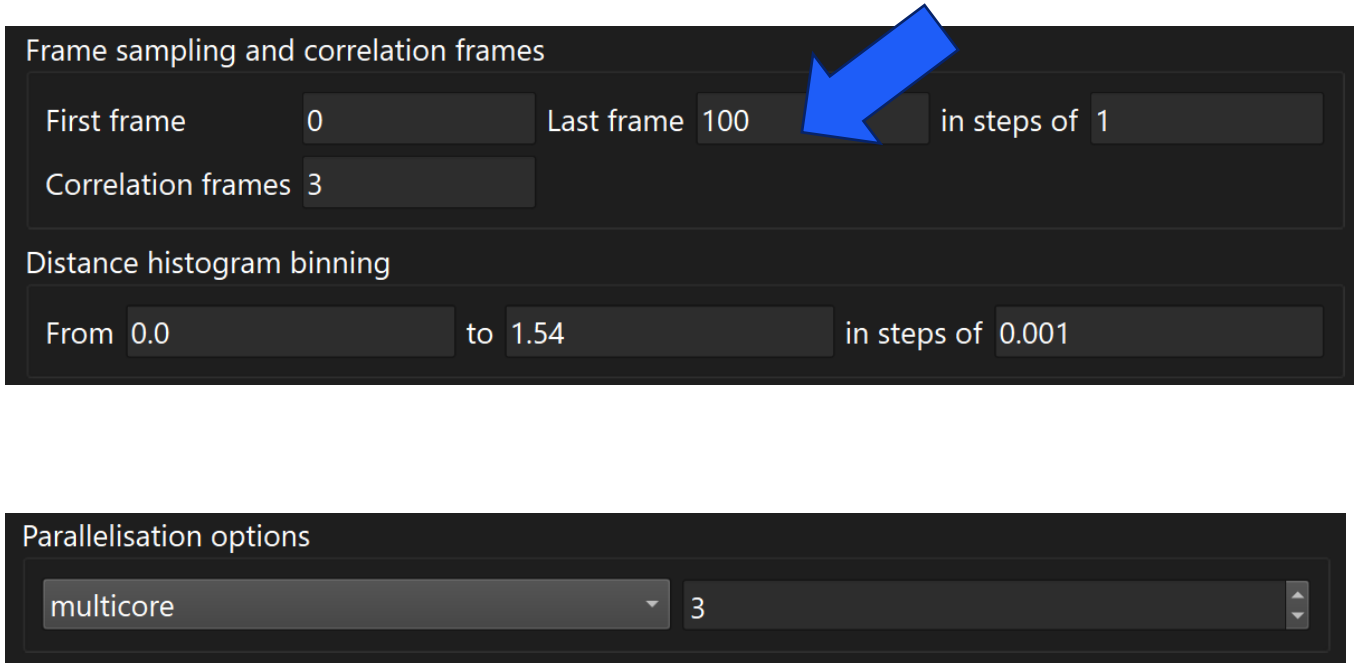
- Now go to the actions tab and select VanHoveFunctionDistinct under analysis and dynamics.
- The VanHoveFunctionDistinct is quite computationally expensive so let's pick specific settings to get the time differences we want.
- We set correlation frames to 3 so that we have results for t=0, 20, and 40 fs.





MDANSE

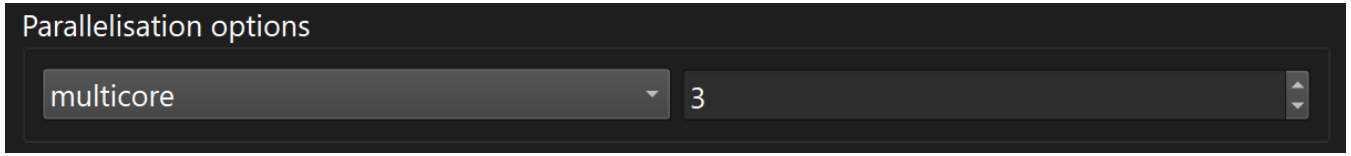
- Now go to the actions tab and select VanHoveFunctionDistinct under analysis and dynamics.
- The VanHoveFunctionDistinct is quite computationally expensive so let's pick specific settings to get the time differences we want.
- We set correlation frames to 3 so that we have results for t=0, 20, and 40 fs.
- We set the in frames of to 100 to reduce the calculation time a bit.



MDANSE

- Here we decrease the in step of setting to increase the resolution of the plots.





MDANSE

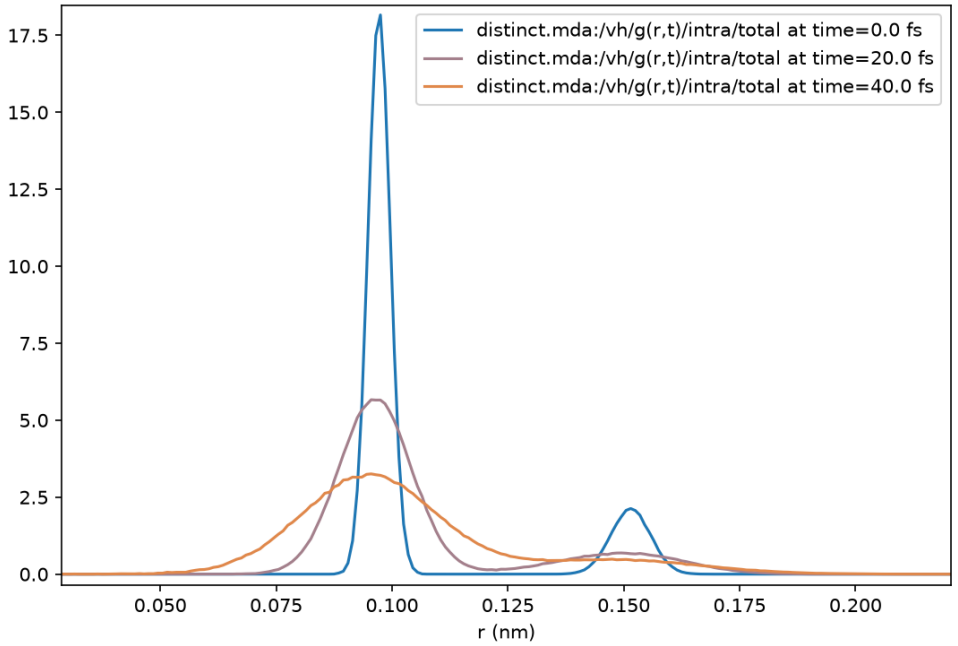
- Here we decrease the in step of setting to increase the resolution of the plots.
- To speed up the calculations you can increase the number of processor used. For van Hove distinct you should set this to be smaller than the number of correlation frames.

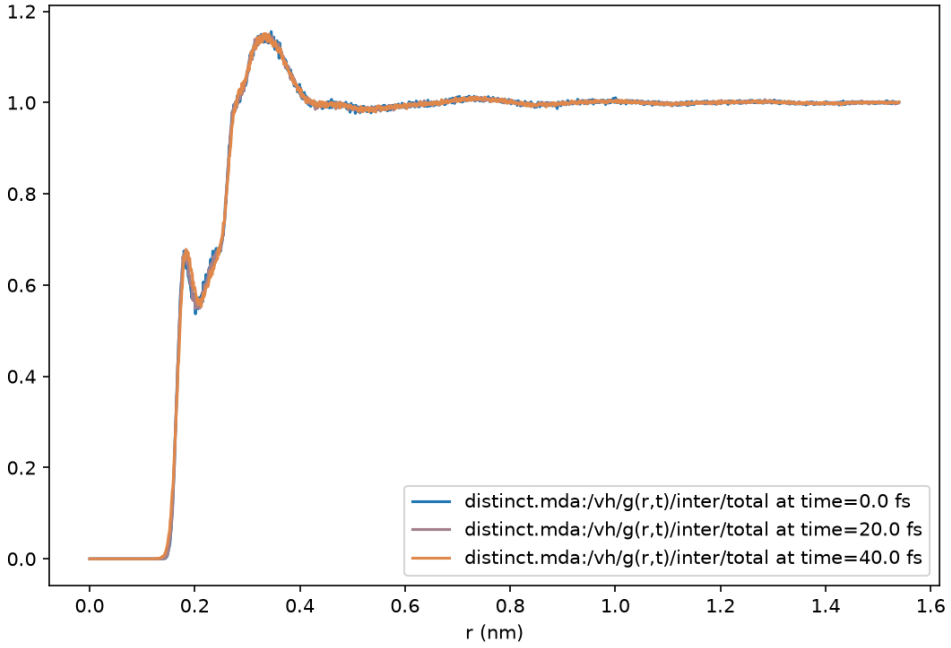




MDANSE

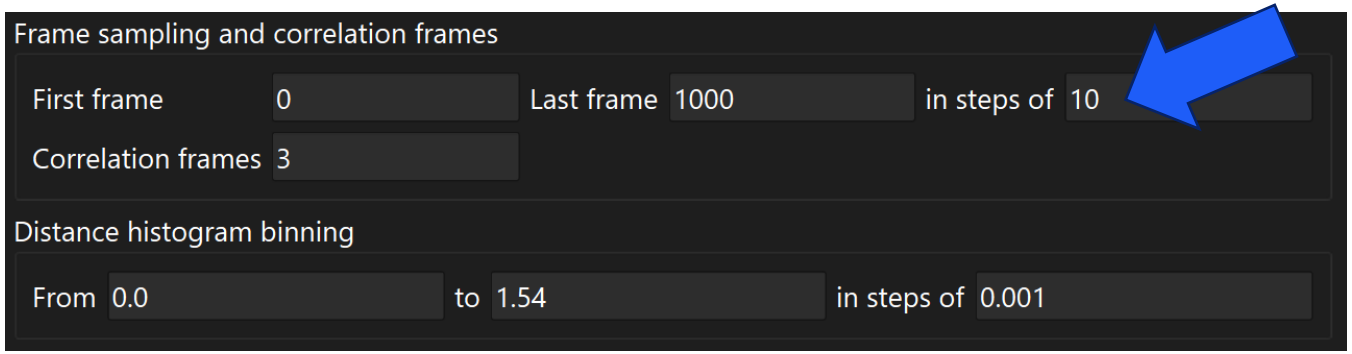
- Go to the plot creator/holder and plot your results. Since there were molecules in the trajectory the results should also be split into both intra and inter molecular parts.
- Have a look at the different results can you explain their behaviours?
- For the intramolecular results can you see which peak belong to which pair of atoms?
- Can you explain why at this time scale the intermolecular results are mostly unchanged while the peaks in the intramolecular results have broaden out?

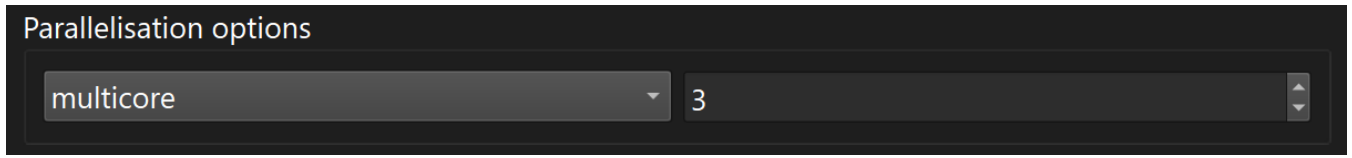




MDANSE

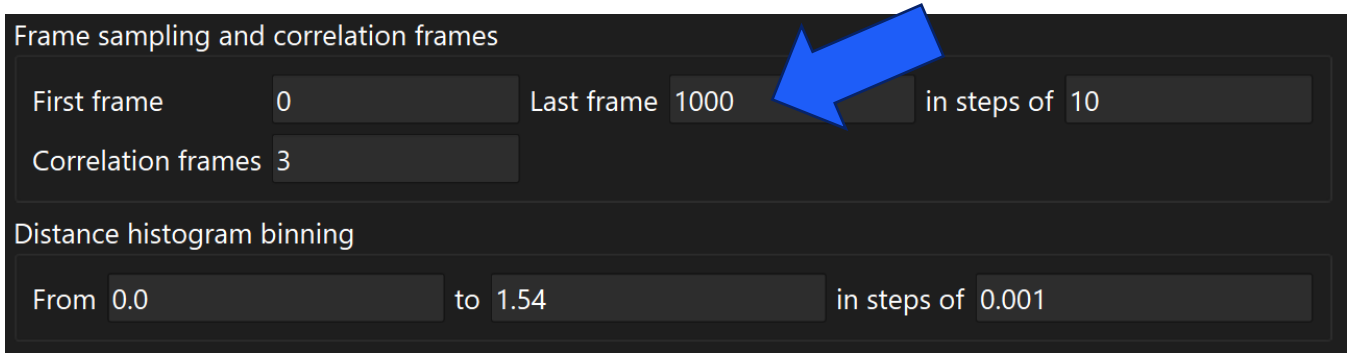
- Let's run the same calculation again but with a larger value for in steps of. This will mean that we will run calculate the van Hove function for times t = 0, 200, and 400 fs.

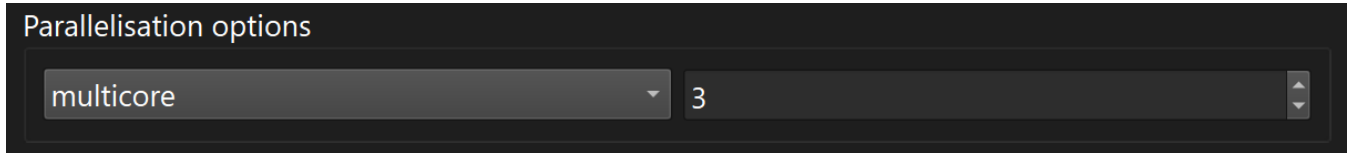




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- Let's run the same calculation again but with a larger value for in steps of. This will mean that we will run calculate the van Hove function for times t = 0, 200, and 400 fs.
- For the last frames set this to 1000.
- Results will a bit noisy, we could use a bigger system with a larger last frame setting, but this would take more time.





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- Plot the results like last time. You can see at this time scale, the intermolecular contribution to the distinct part of the van Hove function begin to change. What is happening at this timescale for this to occur?

